
Lab W1 Submission: Environment Setup

Group submission template

Group members

- [Type name here.]
- [Type name here.]
- [Type name here.]
- [Type name here.]

1 Part A – QEMU checklist

Table 1: Part A checklist

Checklist item	Confirmed by all 4?
QEMU installed and <code>qemu-system-x86_64 -version</code> runs	[Yes/No]
AArch64 cross-compiler installed	[Yes/No]
Native hello-world compiles and runs	[Yes/No]
Cross-compiled AArch64 hello-world runs under QEMU	[Yes/No]

2 Part B – Raspberry Pi output

Paste your `uname -m` output below.

1 [Paste your `uname -m` output here.]

Paste the relevant line(s) from `cat /proc/cpuinfo` below (the `model name` or `Hardware` field is enough, you do not need the whole file).

1 [Paste your `/proc/cpuinfo` output here.]

3 Screenshots

[If you have not added your screenshot files yet, leave a short note here about what you still need to attach.]

4 Part C – Reflection

1. What did `uname -m` report on the Raspberry Pi, and what does that string actually name?
[Type your answer here.]
2. You cross-compiled for AArch64 on your laptop in Part A but compiled natively on the Pi in Part B. In your own words, why does a cross-compiler exist at all? What problem does it solve that a regular compiler cannot?

[Type your answer here.]

3. QEMU let you run an AArch64 binary on an x86-64 laptop. Based on what you observed, what do you think QEMU is actually doing to make that possible?

[Type your answer here.]

Anything that did not work

If any step did not work for someone in the group, describe what you tried here. If everything worked, you can leave this section as-is or delete it.

[Type your answer here, or delete this section.]